

Day 10 Dec 12th 2025

1) Recursion function

A recursive function is a function that calls itself until a base condition is met.

Base case: condition to stop recursion

Recursive case: function call itself with modified input

```
Ex:- def fact(n):  
    if n == 1:  
        return 1  
    else:  
        return n * fact(n-1)  
fact(5)
```

2) Nested function

- * It's a function defined inside another function
- * It helps in encapsulation.

Syntax:-

```
def outer_fun(P1, P2, ..., Pn):  
    def inner_fun(P1, P2, ..., Pn):  
        return statement  
    return inner_fun()  
print(outer_fun())
```


Ex:- def power(exp):
def inner(base):
return base ** exp
return inner
cube = power(3)
print(cube(2))